Rızgar Ozan

Software engineer · games, tools and AI retrieval

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Software engineer in training who builds systems and the tools that prove they work. This summer our team shipped a local LLM stack, FastAPI/Docker data services and an Active Directory-backed internal portal at Barko Elektronik. On my own: a match-3 workbench with an engine-independent rules core, a Level Editor and a bot-driven difficulty curve (12,000 games in ~5 s), a Unity package (OpenUPM) that audits missing TextMeshPro glyphs across a whole project, and a reproducible Turkish retrieval benchmark. Comfortable in C# and Python; tests and measurements before claims.

EDUCATION

Hacettepe University, Ankara — B.Sc. Computer Education and Instructional Technology (CEIT) — Expected graduation June 2027 · GPA 3.02 / 4.00

EXPERIENCE

Software Engineering Intern · Barko Elektronik, Ankara — Jul 2026 – Aug 2026

- Two-person intern team. Deployed a fully local LLM stack on company hardware: benchmarked 7 open-weight models per use case and rolled out a company-wide assistant via Open WebUI with LDAP sign-in.
- Built FastAPI services, containerised with Docker, that pull data from Oracle and feed live Power BI dashboards, replacing manual reporting.
- Applied the local LLM to business workflows: CV analysis and categorisation, inconsistency detection in historical records, department-specific assistants.
- Unified department web apps into one internal portal on a shared design standard; integrated it with Active Directory over LDAP and implemented role-based access control.

SELECTED PROJECTS

Match3 Lab — match-3 rules core, level format and bot simulator

C# (.NET, netstandard2.1) · Unity 6 · xUnit · GitHub Actions github.com/RizgarOzan/match3-lab

- Engine-independent, deterministic rules core (PCG32, event stream instead of callbacks) shared by a .NET CLI and a Unity presentation layer; 51 xUnit + 2 PlayMode tests; playable web demo.
- Parallel bot simulator measures level difficulty: 12,000 games in ~5 s on a laptop; difficulty curve, move-budget auto-tuner and CSV export.
- Found and fixed a bot 'foresight' bug (bot cloned the real RNG) that inflated win rates by up to 20 points; documented as an ADR.

turkish-rag-eval — Turkish retrieval evaluation harness

Python · BM25 · sentence-transformers · pytest · GitHub Actions github.com/RizgarOzan/turkish-rag-eval

- Reproducible benchmark: which parts of a RAG pipeline earn their cost in Turkish? 3 chunking strategies × 4 retrievers × 6 embedding models; 300-question gold set (58 human-written, 242 LLM-drafted and cross-labelled by a second model). Dataset public on Hugging Face, proposed as an MTEB task.
- Measured nDCG@10, Recall@5, MRR and CPU P95 latency. On 296 answerable questions: stemmed BM25 0.557, multilingual-e5-base 0.644, hybrid RRF 0.662; a 5-character prefix stemmer alone lifts BM25 by 23–29 % on the human-written set.
- Implemented abstention analysis (coverage vs. selective accuracy) and showed the top-1 margin on fused scores carries no signal; raw cosine does.

tmp-glyph-audit — Unity package that finds every TextMeshPro text your fonts cannot draw

C# · Unity Editor · UPM · OpenUPM · xUnit github.com/RizgarOzan/tmp-glyph-audit

- Walks scenes, prefabs and runtime text through TMP's real fallback chain and reports missing glyphs per text/object/character; editor window plus CLI with CI exit codes.
- Pure C# core with 24 tests + 5 EditMode tests; published on OpenUPM (v0.2.0).

lockstep-tactics — deterministic multiplayer tactics game

C++20 · WebAssembly · WebRTC · GitHub Actions github.com/RizgarOzan/lockstep-tactics

- Deterministic lockstep simulation that two browsers run in sync over WebRTC; a CI job checks bit-identical state hashes on MSVC, Apple clang and WebAssembly on every push.
- Rule-based AI that beats a random player in 998 of 1,000 matches and a kiting script 580 to 384.

Code:Enigma — personalised math learning web app (2-person team)

React 19 · Vite · Tailwind · Node/Express

- Built the React dashboard pages (learning-gaps view with filters, settings, peer averages and progress marker) on a React Router app backed by a Node/Express API that serves LLM tutor prompts.

OPEN SOURCE

Open-source contributions — 156 merged pull requests to 40 repositories I do not own (2026)

- ML / AI: embeddings-benchmark/mteb (23) and results (5), kornia/kornia (18: numerical edge cases in polynomial solvers and geometry, e.g. NaN on float16 or degenerate roots), HKUDS/LightRAG (4), docling (8), huggingface/sentence-transformers (4). Unity: CoplayDev/unity-mcp (9).
- Also mdn, radzen-blazor, atlassian-python-api, manim, pylint, openupm and others; 14 more are open and in review. Every PR reproduced first, with a regression test and full local CI.

SKILLS

Languages	Python, C#, C++20, TypeScript, JavaScript, SQL, HTML/CSS
Full stack	React (Code:Enigma), Node.js and Express (Code:Enigma API), MongoDB and Supabase (coursework), REST APIs with FastAPI
Engines & tools	FastAPI, Docker, Oracle / SQL, LDAP / Active Directory, Unity 6 (Editor tooling, UPM), .NET, xUnit, pytest, Git/GitHub Actions
AI / data	Local LLM deployment (Open WebUI), retrieval evaluation (nDCG, MRR), BM25 / dense / hybrid retrieval, RAG, sentence-transformers, Power BI
Spoken	Turkish (native), English (B2, working proficiency)
Also shipped	**Bilim Dedektifi**: released educational detective game, team course project at Hacettepe (Unity, C#). rizgarozan.itch.io/bilim-dedektifi